1. What is CSS? Explain the full form of CSS

CSS is a stylesheet language used to control the apperance and layout of HTML elements. HTML creates structure of a webpage, while CSS makes the webpage beautiful, responsive, and visually attractive

2. Cascading means flowing down according to priority. When multiple CSS rules target the same element, the browser decides which rule should be applied based on the CSS cascade rules

Ex:

```
h4 { colour : black; }
h4 { color : white; }
```

<u>Output</u> :-

white, Because the last rule overrides the 1st one.

Real-world examples: Shopping websites

3.

Cascading	Specificity
* Cascading is the overall process CSS uses to decide which style should be applied when multiple rules conflict	* Specificity is the priority / weight of a selector
* It considers importance, specificity and source order	It compares selectors such as ID, class and element selectors
* It decides which CSS declaration wins	It helps determine which selector is stronger

4. Three types of CSS
   * Inline CSS
   * Internal CSS
   * External CSS

In this External CSS is recommended because CSS stored inside a separate file and it is Reusable, Professional, Faster Development, Easy Maintainance

5) i) Universal Selector

⇒ Selects every element

Syntax

```
*{
margin:0;
padding:0;
}
```

ii) Element Selector

⇒ Targets HTML tags

Example:-

```
P{
color: red;
}
```

iii) Class Selector

⇒ Starts with .

Ex:-

```
.title{
color: red;
}
```

iv) ID Selector

⇒ Starts with #

HTML

```
<h1 id="logo'>
```

CSS

```
#logo{ color: black;
}
```

(v) Group Selector

⇒ Multiple selectors

Ex:-

```
P, h1, div{
white, yellow;
}
```

(vi) Descendant Selector

⇒ Targets children inside another element.

```
div P{
color: red;
}
```

(vii) child Selector

⇒ Direct child

```
div> P{
color: blue;
}
```

6)

Class Selector	ID Selector
* Used to style multiple elements	Usually used to style one unique element
* using a .	using a #
⇒ Ex:- .student	Ex: #student
* A class can be used many times on a page	An ID should be unique within a page.

7)

Descendant Selector	~~ID~~ child Selector
Uses a space	Uses >
Selects elements at any level inside another element	Selects only direct child
Ex:- div p	Ex: div > p

8. CSS Box Model is the fundamental concept used to understand how every HTML element occupies space on a webpage
Every HTML element is treated as a rectangular box by the browser whether its p, h1, div, img etc
The box model consists of 4 parts
1. Content 2. Padding 3. Border 4. Margin

9.

Padding	Margin
Padding is the space between the content and the border.	Margin is the space outside the border.
Padding increases the internal spacing	Margin separates one element from another

10. box-sizing is to controls how width and height are calculated
Recommended: box-sizing: border-box;
Width includes content, padding and border

11.

Block	Inline	Inline - Block
New line	Same line	Same line
Full width	required width	width works
Width works	Width doesn't work	width works
Height works	Height doesn't work	Height works

12. Can we apply width and height to: div, span, button & why
<div> → yes
i) <div> is a block-level element by default we can apply width & height.

E)
```
div{ width: 200px;
     height 100px;
     background color: blue; }
```

<span> → not normally.
ii) Span is an inline element by default. The width and height generally do not affect an inline <span>

```
span{
  width: 200px;
  height: 100px;
}
```

To make them work:

```
span{ display: inline-block;
      width: 200px;
      height: 100px; }
```

⇒ Now width and height work

ii) Justify - content : center;
centers the flex item along the main axis
with the default flex-direction : row, the main axis
is horizontal.
→ Centers horizontally.

iii) Align-items : center;
Centers the flex item along the cross axis.
with the default row direction, the cross axis is
vertical.
→ centers vertically.

iv) height : 100vh;
makes the container as tall as the entire browser
viewport.
vh → viewport height.
100vh = 100% of the screen height.

ii) <button> - yes
button is an inline-block-level element by default, so width and height can be applied

```
button {
    width : 50px;
        height : 50px;
}
```

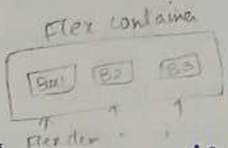
13) Difference b/w display: none; visibility: hidden;
display : none → Hidden & No space.
visibility : hidden → Hidden & Space Reserved

14) Flexbox in CSS, it is a layout system used to arrange elements in a row or column easily

i) Flex container ⇒ There are the four basic concepts of CSS Flexbox
The element where we write

```
.container{
    display: flex;
}
```

is called the flex container.



Box 1, 2, 3 are flex item

ii) Flex items
The direct children of a flex container are called Flex items

iii) Main Axis is the primary direction in which flex items are arranged.
flex-direction : row; (or) flex-direction : column;

iv) Cross Axis is perpendicular to the main axis.
when: flex-direction : row;
* Main axis → direction of flex items
* Cross axis → perpendicular to the main axis

15. How do you perfectly center a div using Flexbox? Write the CSS code and explain each property.

```
.container{
display : flex;
justify-content : center;
Align-items : center
height : 100vh;
}
```

1. display : flex;
makes container a flex container. Its child elements flex items